

When Does Trajectory Prediction Help PC-FMCW/DPSK Vehicular Optical Scheduling?

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Abstract

Phase-coded frequency-modulated continuous-wave (PC-FMCW) laser headlamps can combine sensing, illumination and differential phase-shift-keying communication, but the corresponding optical link is strongly geometry dependent. This work studies a non-Joint communication-control question: whether a scheduler should serve a receiver now because causal trajectory prediction indicates that its link will soon become unusable. We implement a packet-level pipeline from observed motion to trajectory forecasts, ego-relative range and bearing, normalized optical gain, SNR, DBPSK bit and packet errors, link lifetime, queues, deadlines and receiver scheduling. Frame consistency, stationary headings, end-of-record horizons, physical time units, censoring and clustered statistics are covered by 44 regression tests and five scientific gates. In a corrected 12-episode controlled benchmark, Link-Lifetime scheduling changes mean goodput from 2.293 to 2.307 Mbps relative to Reactive Greedy, but the paired bootstrap 95% interval $[-0.0319, 0.0593]$ Mbps includes zero and Holm-adjusted Wilcoxon $p=1.0$. Its P95 latency is 269 ms worse on average. A three-scene compact WOMB proxy benchmark is also negative: 1.056 versus 1.160 Mbps. A 430-row two-seed staged diagnostic shows operating-region sign changes but is insufficient for inference. The current evidence therefore rejects a universal prediction-gain claim and motivates a narrower question: under which link-closure, traffic and uncertainty regimes does future mobility information have communication value? Official WOMB shards and a compatible learned checkpoint are still required for a submission-quality empirical answer.

Index terms: predictive communication, vehicular optical communication, PC-FMCW, DPSK, trajectory forecasting, link lifetime, packet scheduling.

I. Introduction

Integrated sensing, communication and illumination can reuse an optical vehicle front end for ranging, lighting and high-rate data transmission. The supplied Part-A PC-FMCW study provides a physical-layer premise with a 193.4 THz optical carrier, 10 GHz chirp bandwidth, 10 μ s chirp duration and 1 Gbit/s DPSK data rate. It does not define a mobility-aware packet scheduler.

A current-link scheduler sees only the instantaneous communication opportunity. If one receiver is approaching an optical field-of-view boundary, packets left in its queue may soon become undeliverable. A predictor could expose that closing window. Prediction can also harm scheduling: trajectory error can cross a hard optical boundary, an urgency heuristic can sacrifice low-latency service, and traffic saturation can eliminate scheduling flexibility.

This paper asks: when does causal future-motion information improve receiver scheduling for a PC-FMCW/DPSK-informed optical link? The decision is the receiver served in the current slot. Beam-index selection, adaptive driving-beam control and illumination optimization are explicitly excluded.

The contributions are:

- an auditable causal motion-to-link-to-packet pipeline;
- frame- and time-consistent link-aware prediction and scheduling;
- classical, reactive, predictive and perfect-future information references;
- explicit BER/PER/goodput outage semantics and normalized-power boundaries;
- packet censoring, scenario slices and cluster-aware paired inference;
- a true-SDC official-WOMB adapter and versioned learned-checkpoint contract;
- corrected controlled and compact-proxy results, including null and negative findings.

II. System Model

At slot t , the system observes ego and target positions through t , packet queues, deadlines and past service. A deployable predictor estimates the next H positions. Predicted target and ego trajectories are converted to range $d(i,k) = ||\hat{p}(i,t+k) - \hat{p}(e,t+k)||_2$ and wrapped ego-relative bearing $\phi(i,k) = \text{wrap}[\text{atan2}(\Delta_y, \Delta_x) - \psi(e,t+k)]$.

Stationary ego samples retain their last valid heading. Unavailable future steps are truncated for every policy. A deployable predictor never receives a ground-truth sample after t ; future-mutation tests enforce this invariant.

The scheduler selects at most one target. Its forecast may influence the decision, but packet success is sampled from the current ground-truth-derived link. This separates decision information from evaluation and prevents self-confirming results.

III. PC-FMCW/DPSK-Informed Link

Because the supplied files contain no calibrated vehicle-to-vehicle optical budget, the model is anchored by a declared reference SNR rather than absolute measurement. Relative gain combines inverse footprint area, atmospheric loss, Gaussian pointing loss and a hard field of view. In compact notation, $h(d, \phi)$ is proportional to inverse footprint area multiplied by $\exp(-\kappa d)$, Gaussian pointing loss and the indicator $1(|\phi| \leq FOV/2)$.

The SNR is $\gamma_{\text{ref}} h(d, \phi)/h(d_{\text{ref}}, 0)$. The power output is a normalized reference quantity with `received_power_calibrated=false`.

Analytical DBPSK uses $P_b = 0.5 \exp(-\gamma)$. For L packet bits, $PER = 1 - (1 - P_b)^L$, and successful goodput is $R_b(1 - PER)$. A second BER path uses fixed-seed symbol-level Monte Carlo from -5 to 25 dB with adaptive bit counts and a Wilson 95% upper bound when no error is observed.

BER, PER and minimum-goodput outage flags are all stored. The configured mode determines link lifetime, defined as the first predicted outage step, or the censored horizon if no outage is predicted.

IV. Motion, Sensing and Learning

The deterministic baselines are Last Position, Constant Velocity, Constant Acceleration, a position-only Kalman CV filter and a lightweight causal CV/CA IMM. Perfect future motion is an information reference.

The optional GRU can be trained under four objectives: trajectory-only, trajectory+link, trajectory+outage and full. The link loss receives explicit ego heading; the outage loss combines a smooth field-of-view surrogate for gradient flow with exact hard-boundary evaluation. Each local checkpoint records its feature schema, dataset hash, split and seed. The upstream Stage-4 reports use a different eight-feature interface and no compatible checkpoint was supplied.

Robustness studies distinguish future forecast degradation from observation uncertainty. The declared sensing model supports IID Cartesian noise and range/bearing noise with range-dependent radial variance and AR(1) temporal correlation. The Kalman path can use an isotropic-equivalent covariance. These are simulation assumptions, not measured sensor specifications.

Predictor	Synthetic ADE (m)	Synthetic SNR MAE (dB)	Proxy ADE (m)	Proxy lifetime MAE (s)
Last Position	1.711	0.698	1.903	0.117
Constant Velocity	0.087	0.031	0.097	0.056
Kalman CV	0.352	0.083	0.328	0.026
IMM	0.176	0.053	0.212	0.054
Constant Acceleration	0.028	0.002	0.196	0.051
Perfect Future	0.000	0.000	0.000	0.000

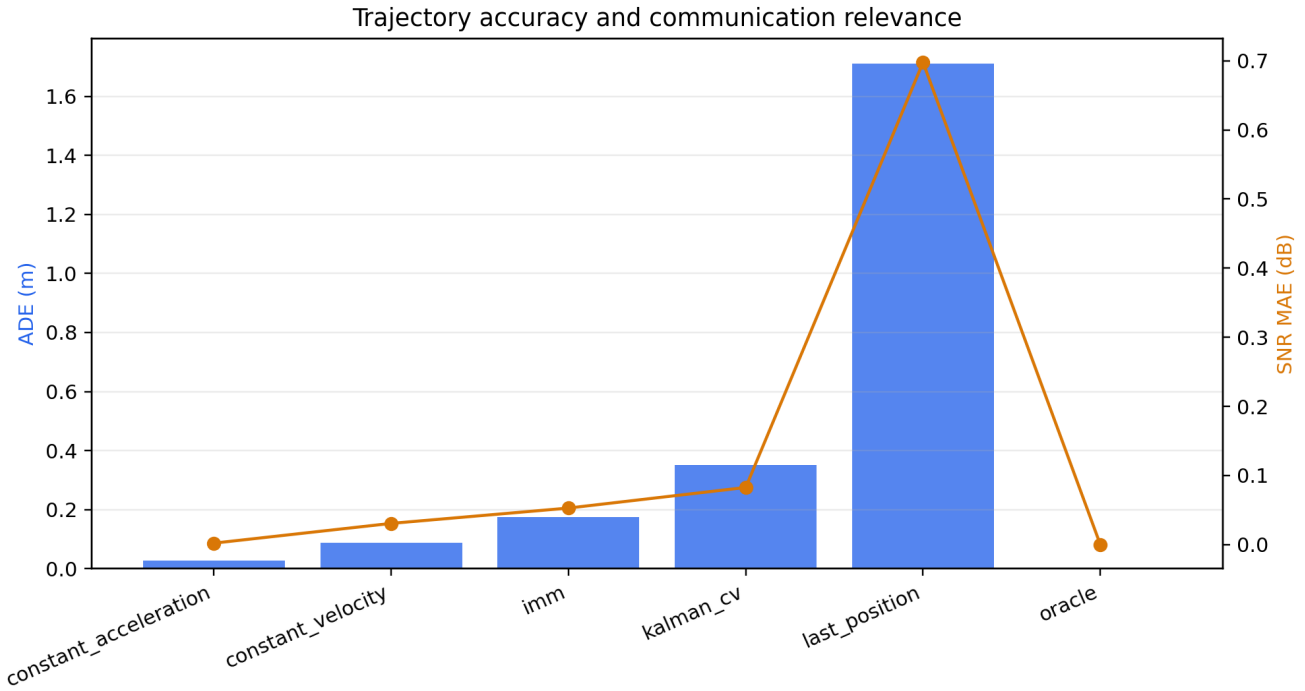


Fig. 1. Motion ADE and derived-link SNR error on controlled trajectories. Link metrics are evaluated separately because a small position error can cross a hard FoV boundary.

V. Traffic and Scheduling

Each target has a bounded FIFO queue. Poisson, periodic, Markov-modulated and saturated arrivals are supported. Deadlines and episode duration are expressed in physical seconds and converted only after slot duration is chosen. Failed packets return to the queue. Packets remaining at termination are censored and reported explicitly.

Reactive references are Random, Round Robin, Reactive Greedy and Proportional Fair. Predictive policies apply CV, Kalman, IMM or constant-acceleration forecasts to a utility containing discounted future goodput, outage, queue, deadline, fairness, opportunity and switching terms.

Link Lifetime adds $U_{\text{life}}(i) = w_{\text{life}} \text{normalized_queue}(i) \max[0, 1 - T_{\text{link}}(i)/H]$ only when the current link is usable. The term is dimensionless and invariant to proportional rescaling of horizon steps. The Oracle receives perfect future positions but uses the same heuristic; it is not a global scheduling optimum.

VI. Experimental Protocol

The controlled benchmark has 12 independent 12 s episodes, five receivers, 100 ms slots, 1 s horizon, load 0.72 and packet deadlines of 1.2 $\pm$ 0.4 s. All policies share motion seeds, arrivals, deadlines and per-attempt random draws.

The compact export contains 56 actors and three WOMB scenario IDs with one second of history and one second of future. Because it omits the official SDC, the adapter uses a deterministic medoid proxy ego. The communication link is simulated.

A staged design changes one axis around the reference: load, physical horizon, slot duration, receiver count, traffic model, packet size, deadline, reference SNR, FoV, outage mode or sensing model. Its quick run uses two seeds and is diagnostic. Final inference requires five seeds and official held-out scenes.

Metrics include goodput, PDR, availability and scheduled outage, P50/P95/P99 latency, deadline miss, censoring, delivered-before-expiry, queue at first disconnection, demand-normalized fairness, and scheduled SNR/BER/PER. Paired statistics operate at independent scenario/seed level, use metric-specific direction and apply Holm correction.

VII. Corrected Results

Policy	Goodput (Mbps)	PDR	P95 latency (ms)	Deadline+censored	Demand Jain
Reactive Greedy	2.2930	0.6439	699.6	0.3561	0.6717
CV Predictive	2.3053	0.6473	948.7	0.3527	0.6962
Kalman Predictive	2.3047	0.6472	933.3	0.3528	0.6948
IMM Predictive	2.3073	0.6479	948.7	0.3521	0.6949
Predictive Utility	2.3063	0.6476	965.4	0.3524	0.6972
Link Lifetime	2.3070	0.6478	968.3	0.3522	0.6993
Oracle Information	2.3078	0.6480	968.3	0.3520	0.7000

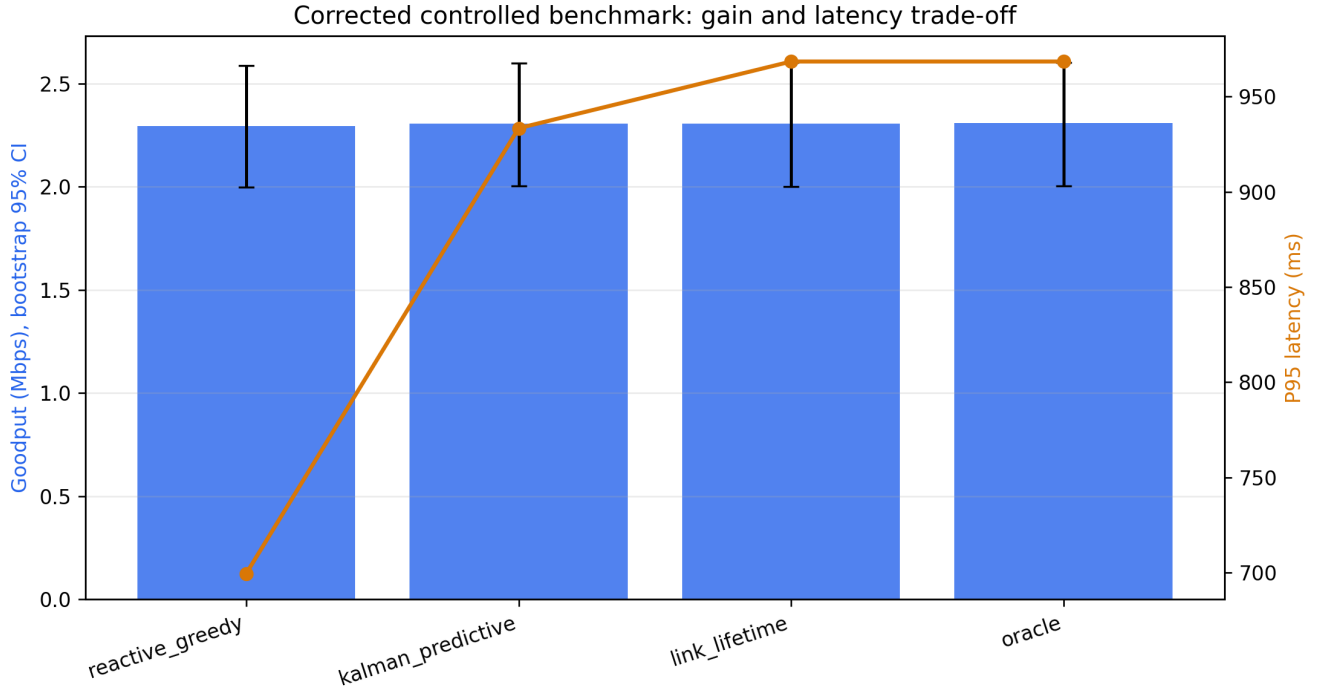


Fig. 2. Corrected controlled benchmark. Predictive policies show slightly higher mean goodput and fairness but substantially worse P95 latency.

Link Lifetime minus Reactive goodput is +0.0140 Mbps with paired bootstrap 95% interval [-0.0319, 0.0593], win fraction 66.7%, Cohen $d_z=0.161$, Wilcoxon $p=0.677$ and Holm-adjusted $p=1.0$. The data do not establish a goodput gain.

The P95 difference is +268.8 ms, unfavorable. Its paired interval is entirely positive; Holm-adjusted Wilcoxon $p=0.00439$. Thus the clearest controlled effect is a latency cost, not a throughput improvement.

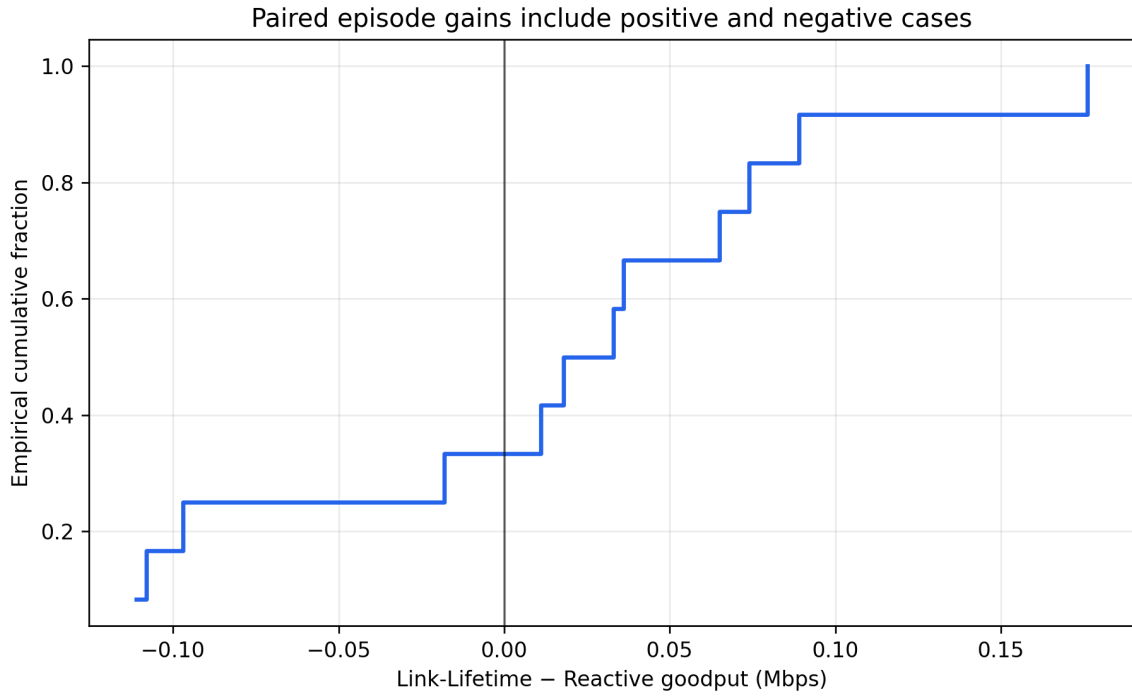


Fig. 3. Episode-level *Link-Lifetime* minus *Reactive goodput* differences include both positive and negative outcomes.

In the quick lifetime ablation, Predictive Utility, Link Lifetime and Link Lifetime with zero lifetime weight all obtain 2.1655 Mbps. The explicit lifetime term is inactive in those default episodes because it does not change a decision inside the available closure horizon. This null result prevents the earlier interpretation that lifetime urgency explained a default gain.

The compact WOMD proxy result is also negative: Reactive obtains 1.160 Mbps and 0.329 PDR, while Link Lifetime obtains 1.056 Mbps and 0.299 PDR. Approximately 70% of packets are deadline-dropped or censored in the one-second evaluation window. The three scenes cannot establish generalization.

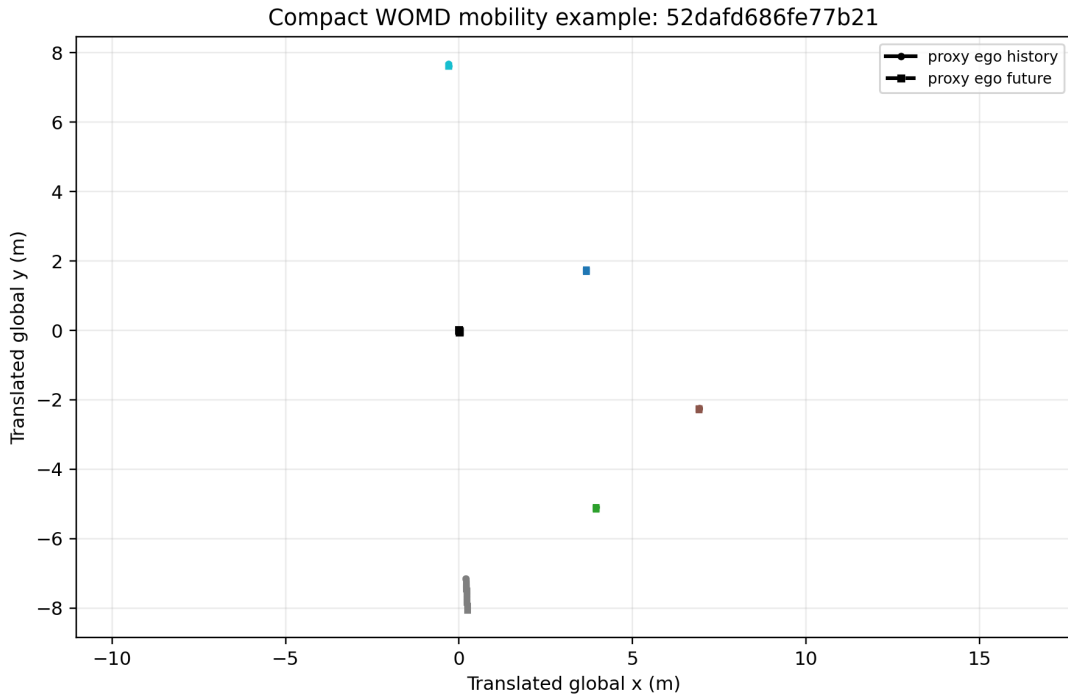


Fig. 4. Compact real-motion example. The proxy ego is an integration device, not the official SDC identity.

The 430-row staged quick run contains 43 settings, two seeds and five policies. The Link-Lifetime minus Reactive difference changes sign across axes - for example, +0.018 Mbps at load 0.3 and -0.172 Mbps at load 0.9. Such changes are hypothesis-generating only because two seeds do not identify an effect.

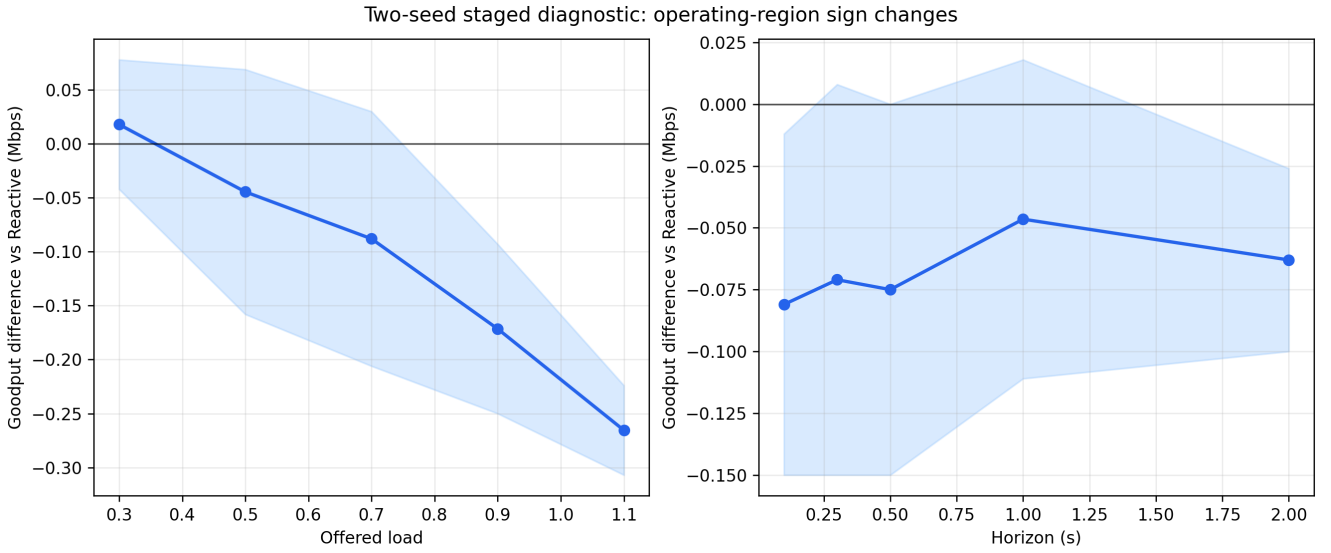


Fig. 5. Two-seed staged diagnostic. Shading spans the two seed outcomes, not a publication confidence interval.

VIII. Limitations

WOMD provides motion, not optical measurements. Every link and packet outcome is model-based. The normalized power reference cannot support an absolute watt claim. The compact export lacks true SDC identity, raw validity masks, map context and a long evaluation window.

The official-WOMD adapter and learned ablation runner are implemented, but raw WOMD shards, a compatible checkpoint and PyTorch were unavailable in the executed environment. No learned result is reported. The supplied Stage-4 JSON reports are provenance rather than locally reproduced evidence.

The Oracle is only an information reference and can be worse than Reactive because the heuristic is not an exact offline optimum. A small-instance exact optimizer remains future work. Channel parameters, traffic and scheduler weights require development-set freezing before final testing.

IX. Conclusion

The corrected codebase demonstrates how causal mobility forecasts can be translated into optical-link forecasts and packet scheduling decisions. It also shows why a favorable controlled result cannot be assumed: the mean goodput difference is uncertain, tail latency is worse, the explicit lifetime ablation is null at the default point and the compact proxy benchmark is negative.

The scientifically useful next question is conditional rather than universal: which link-closure geometries, deadline pressures and uncertainty levels create enough scheduling flexibility for prediction to help? Answering it requires the official WOMD held-out evaluation and a real multi-seed learned-model ablation.

References

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